

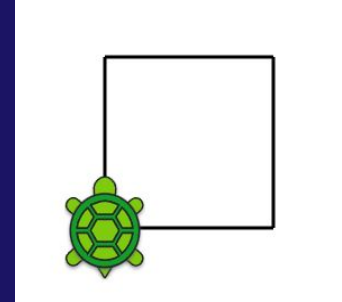


Recap & Recall

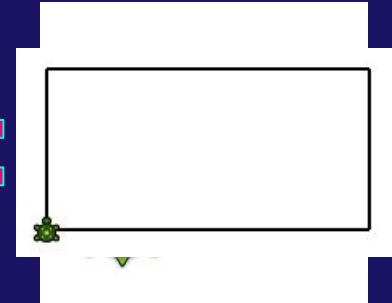
In Small Basic, the code below draws a square on the screen.

On your MWB, draw the shape this code will create.

```
Turtle.Move(100)
Turtle.Turn(90)
Turtle.Move(100)
Turtle.Turn(90)
Turtle.Move(100)
Turtle.Turn(90)
Turtle.Move(100)
Turtle.Turn(90)
```



```
Turtle.Move(100)
Turtle.Turn(90)
Turtle.Move(200)
Turtle.Turn(90)
Turtle.Move(100)
Turtle.Turn(90)
Turtle.Move(200)
Turtle.Turn(90)
```



Lesson Question

How do I output data?

Topic Question

How do I program a computer?



PYTHON PROGRAMMING

Y8 Python Programming

Lesson Question	How do I output data?
Topic Question	How do I program a computer?

Y8 - Programming - Python



How do I program a computer?

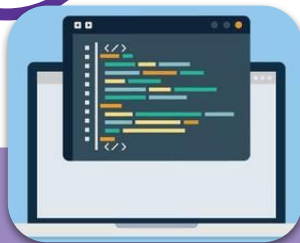


**Outwood
Travels**

Assessment

Feedback

How can I
create a simple
calculator in
Python?

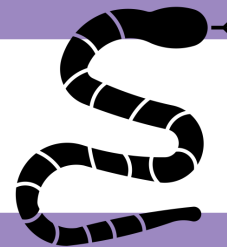


What code lets a computer
perform a calculation?

How can a computer respond
to what I tell it to do?

How can a computer
make decisions?

Where can I store
data in a program?



Theory

How do I
output data?

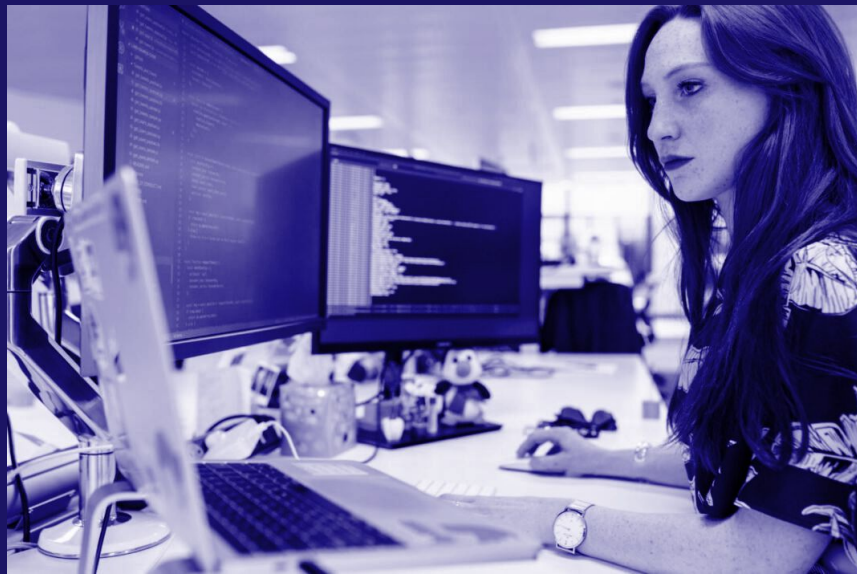




Python Programming

Over the next few weeks you are going to learn how to create different computer programs in Python.

Python is a programming language used in the real world to solve complicated problems.



Lesson Question

How do I output data?

Topic Question

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Getting Started

The **print** command displays text onto the screen.

```
print("The text in green gets printed on the screen")
```



The text in green gets printed on the screen

Discuss in your row what will the following code do? Record your answer on a whiteboard

```
print("You can use more than one print command")  
print("To write text across different lines")
```

Lesson Question	How do I output data?
Topic Question	How do I program a computer?



Task 1a

```
print("The text in green gets printed on the screen")
```



The text in green gets printed on the screen

Open 1. Input & Output - Student Evidence - Task 1

Record your own response in the box

I think the code will...

Lesson Question	How do I output data?
Topic Question	How do I program a computer?



Run Your Code

Open 1. Input & Output - Student Evidence - Task 1b

Copy the code from slide 1 and hit 'run'



```
print("You can use more than one print command")
print("To write text across different lines")
```

Lesson Question	How do I output data?
Topic Question	How do I program a computer?



Investigate Your Code

- Go to **task 1c** (slide 2).
- Answer the questions about the code you have just run. You have ~4 minutes
- Make sure you are giving detail in your answer, and typing in full sentences.

Investigate

Task 2

Answer the following questions:

Question	Type your answer here. Do not give one or two word answers.
Was your prediction correct? What happened when you ran your code?	
What happens when you change print to Print? (i.e. use a capital P)	
What happens if you remove the brackets? (i.e. print "You can...")	
What happens if you remove the speech marks at the end of a line?	
Extension: Search on Google - what is a multi-line print in Python? How can it be used?	

Lesson Question**How do I output data?****Topic Question****How do I program a computer?**



Investigate Your Code

What were your answers to each question?

Investigate 

Task 2

Answer the following questions:

Question	Type your answer here. Do not give one or two word answers.
Was your prediction correct? What happened when you ran your code?	
What happens when you change print to Print? (i.e. use a capital P)	
What happens if you remove the brackets? (i.e. print "You can...")	
What happens if you remove the speech marks at the end of a line?	
Extension: Search on Google - what is a multi-line print in Python? How can it be used?	

Lesson Question	How do I output data?
Topic Question	How do I program a computer?



Modify Your Code

Go to **task 1d** (slide 3). You are now going to modify the code that you have been given to display the following when run:

```
I can modify my code to say  
anything I want on the screen
```

Take a screenshot after each modification and put it into your evidence diary.

Lesson Question	How do I output data?
Topic Question	How do I program a computer?



Make Something New

Go to **task 1e** (slide 4). Make a new program that displays your name and the names of four of your friends on different lines.

Once finished:

- Use a multi-line print command to do this
- Sort the list alphabetically. How have you done this? How might a computer do this automatically?

Lesson Question	How do I output data?
Topic Question	How do I program a computer?



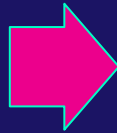
Data Types

There are different data types in Python.

Strings - Are text. They have speech marks around them. They will display exactly as they are shown in the speech marks.

```
print("This is a string")
```

```
print("2 + 2")
```



```
This is a string
```

```
2 + 2
```

Integers - Are whole numbers. Python will try to calculate any maths involving integers.

```
print(2 + 2)
```



```
4
```

Lesson Question	How do I output data?
Topic Question	How do I program a computer?



Predict

Discuss in your row - what will get displayed onto the screen?

```
print("1 + 5")  
  
print(1 + 5)  
  
print("One plus five equals", 1 + 5 )
```

Lesson Question	How do I output data?
Topic Question	How do I program a computer?



Task 2a

Go to **task 2a** (slide 5) of your student workbook

Record your own response in the box

I think the code will...



Lesson Question	How do I output data?
Topic Question	How do I program a computer?



Run Your Code

Go to **task 2b** (slide 5), copy and run the code.

```
print("1 + 5")  
print(1 + 5)  
print("One plus five equals", 1 + 5)  
print("Two times five equals", 2 * 5)
```

Lesson Question	How do I output data?
Topic Question	How do I program a computer?



Investigate Your Code

- Go to **task 2c** (slide 6).
- Answer the questions about the code you have just run. You have ~5 minutes
- Make sure you are giving detail in your answer, and typing in full sentences.

Investigate

Task 6

Answer the following questions:

Question	Type your answer here. Do not give one or two word answers.
Was your prediction correct? What happened when you ran your code?	
What does the symbol * mean? What could / mean?	
Which line is using a string, and which is using integers?	
Do the blank lines between the print commands matter? What happens when you remove them?	
Extension: What other data types are there in Python? Can you show how to use a float instead of an integer?	

Lesson Question	How do I output data?
Topic Question	How do I program a computer?



Investigate Your Code

What were your answers to each question?

Investigate



Task 6



Answer the following questions:

Question	Type your answer here. Do not give one or two word answers.
Was your prediction correct? What happened when you ran your code?	
What does the symbol * mean? What could / mean?	
Which line is using a string, and which is using integers?	
Do the blank lines between the print commands matter? What happens when you remove them?	
Extension: What other data types are there in Python? Can you show how to use a float instead of an integer?	

Lesson Question	How do I output data?
Topic Question	How do I program a computer?



Modify Your Code

Got to **task 2d** (slide 7). You are now going to **calculate** 9-4 and then display both the result and the calculation as a string.

Take a screenshot after each modification and put it into your evidence diary.

Lesson Question	How do I output data?
Topic Question	How do I program a computer?



Make Something New

Go to **task 2e** (slide 8). Make a maths quiz that asks three different questions to the user. You can display the answers after a delay.

It should then output the result of this calculation with a suitable message.

Lesson Question	How do I output data?
Topic Question	How do I program a computer?



Plenary

Write into your planners that Python Homework 1 is due **next lesson**.

Then on your MWB, write a line of code that will print your name onto the screen.

Lesson Question	How do I output data?
Topic Question	How do I program a computer?